

Supplementary Material for “A Two-Phase Particle Swarm and Variable Neighborhood Search Algorithm for Continuous Wind Farm Layout Optimization”

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Tables, figures, sections, algorithms and equations of this document are numbered S1, S2, $\dots$; references without the prefix S (e.g., Table I or (5)) point to the main text. Unless stated otherwise, all runs use 30 seed-paired runs of 6,030 objective calls, and the method names are those of the main text (PSO: constriction coefficients (8)).

S-I Model Details

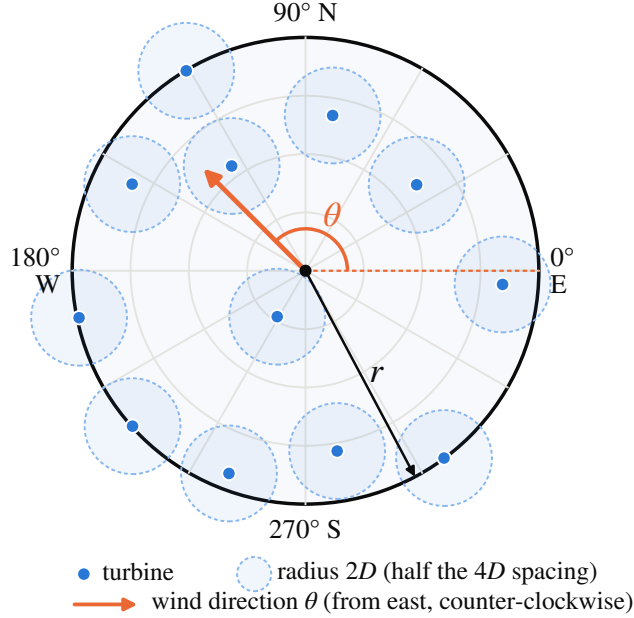


Figure S1: Circular wind farm of radius r . The wind direction θ (direction the wind blows towards) is measured counter-clockwise from east. The points show an optimized layout for 12 turbines in the 750-m farm (Wind Data Set I); the shaded discs have radius $2D$, so two discs touch when their turbines are exactly $4D$ apart.

S-I.A Discrete Form of the Expected Power

Following [1], the integrals of (4) are evaluated with a Riemann sum. The wind direction is divided into bins $\theta_0 = 0^\circ, \theta_1, \dots, \theta_{N_\theta} = 360^\circ$ with probabilities p_l , and the speed range between $s_{\text{cut-in}}$ and s_{rated} into bins $s_0 = s_{\text{cut-in}}, s_1, \dots, s_{N_s} = s_{\text{rated}}$. With $\bar{\theta}_l = (\theta_{l-1} + \theta_l)/2$ and $W_i(s, \theta) = \exp\{-[s/\psi_i(\theta)]^{\zeta(\theta)}\}$,

$$E(P_i) = \sum_{l=1}^{N_\theta} (\theta_l - \theta_{l-1}) p_l \left\{ P_{\text{rated}} W_i(s_{\text{rated}}, \bar{\theta}_l) + \lambda_1 \sum_{q=1}^{N_s} \frac{s_{q-1} + s_q}{2} [W_i(s_{q-1}, \bar{\theta}_l) - W_i(s_q, \bar{\theta}_l)] \right. \\ \left. + \lambda_0 [W_i(s_{\text{cut-in}}, \bar{\theta}_l) - W_i(s_{\text{rated}}, \bar{\theta}_l)] \right\}. \quad (\text{S1})$$

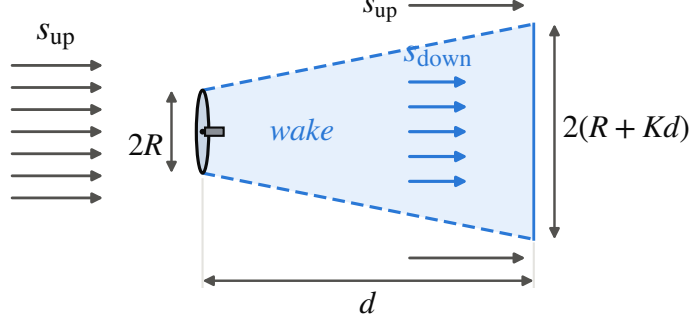


Figure S2: Jensen wake behind a rotor of radius R (schematic, not to scale). The wake widens linearly with the spreading constant K , so that its width at the downstream distance d is $2(R + Kd)$, and the wind speed inside it drops from s_{up} to s_{down} .

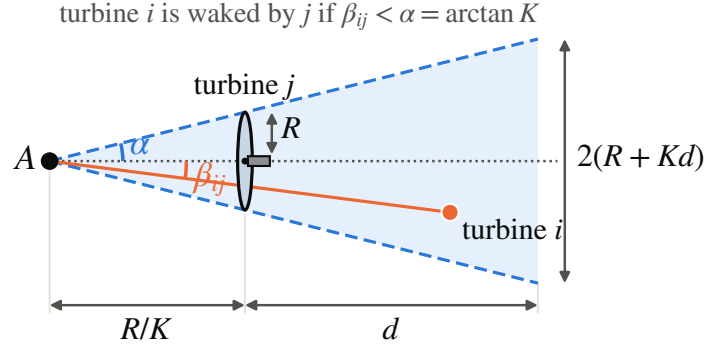


Figure S3: Wake half cone of turbine j (schematic, K exaggerated). The virtual vertex A lies a distance R/K upstream of the rotor, the half-cone angle is $\alpha = \arctan K$, and a downstream turbine i lies in the wake of j when $\beta_{ij} < \alpha$.

Like the earlier studies, we keep the bin-width factor $(\theta_l - \theta_{l-1})$. Because all 24 direction bins are 15° wide and $\sum_l p_l = 1$, the objective equals 15 times the expected farm power in kW, and $\text{AEP} [\text{MWh}] = (\text{objective}/15) \times 8.76$. The linear part of the power curve is -7.0 kW at 3.5 m/s and zero at 3.55 m/s.

S-I.B Gaussian Wake Model Used for the Re-Evaluation

For the re-evaluation of Section VIII-A, the deficit of the Jensen model is replaced by the Gaussian wake model of Bastankhah and Porté-Agel [7],

$$\Delta_{ij} = \left(1 - \sqrt{1 - C_T/(8\sigma^2/D^2)}\right) \exp(-v_{ij}^2/(2\sigma^2)), \quad \sigma/D = K_G d_{ij}/D + \epsilon, \quad (S2)$$

$$\epsilon = 0.2 \sqrt{\frac{1}{2}(1 + \sqrt{1 - C_T})/\sqrt{1 - C_T}},$$

for turbines i downstream of j , with $K_G = 0.04$, the lateral offset v_{ij} and downstream distance d_{ij} of Section II, and the same root-sum-square superposition and Weibull scaling as the benchmark.

S-II Salp Swarm Algorithm and Its Laplacian Variant

Algorithm S1 summarizes SSA [5] and LX-SSA [6] as used in Phase 1 of SSA-VNS and LX-SSA-VNS and as stand-alone methods (Section III-B). A follower moves to the midpoint $\frac{1}{2}(X_m^i + X_m^{i-1})$ between itself and the preceding salp; in LX-SSA it also forms the Laplace candidate (11) and keeps the better of the two, so that one LX-SSA iteration costs $2N_p$ objective calls.

Algorithm S1 SSA and LX-SSA (lines marked LX: LX-SSA only)

```
1: Draw and evaluate  $N_p$  salps; sort them and set the best as food source H.
2: for  $t = 1$  to  $T$  do
3:   Update  $\xi_1 = 2 \exp[-(4t/T)^2]$ .
4:   for salp  $i = 1, \dots, N_p$  do
5:     if  $i \leq N_p/2$  then
6:       Update leader  $i$  using (10).
7:     else
8:       Form the follower candidate  $\frac{1}{2}(\mathbf{X}^i + \mathbf{X}^{i-1})$ .
9:       LX: form  $\mathbf{Y}^i$  using (11), evaluate both candidates and keep the better one.
10:    end if
11:  end for
12:  Apply box repair, evaluate  $F_p$  for the population, sort it, and update H.
13: end for
14: return H.
```

Table S1: Data Set I, 500-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28086.7 (4.5)
3	42137.2	42131.3 (8.5)	42128.7 (8.8)	42128.5 (10.7)	42114.9 (10.3)	42114.8 (14.2)	42130.7 (8.8)	42131.6 (9.7)	42119.9 (11.0)
4	56182.9	56114.4 (20.7)	56110.0 (25.6)	56101.8 (25.3)	56086.8 (19.8)	56078.8 (22.6)	56092.8 (29.8)	56106.1 (21.7)	56057.0 (55.8)
5	70228.7	70051.2 (28.5)	70053.1 (29.3)	69967.2 (159.2)	69994.0 (34.0)	69972.6 (72.0)	69960.5 (78.6)	69876.9 (199.8)	69876.4 (139.3)
6	84274.4	83918.5 (66.9)	83925.6 (55.8)	83780.8 (213.2)	83535.6 (372.5)	83544.9 (465.3)	83326.7 (761.5)	83299.3 (524.5)	83636.4 (227.9)
7	98320.2	97588.7 (292.3)	97569.6 (325.2)	96529.9 (794.9)	96757.0 (692.4)	96380.6 (1013.8)	95903.3 (1316.3) ¹⁵	96967.1 (590.6)	97049.0 (477.4)
8	112365.9	110777.2 (718.5)	110957.4 (615.4)	109316.4 (1117.4)	109172.6 (1362.4)	108258.3 (1704.3) ²⁸	107632.7 (1592.8) ²	109493.5 (2035.2)	110233.3 (571.9)
9	126411.6	123153.2 (1132.3)	123499.5 (1006.4)	121985.3 (1201.8)	120043.1 (2755.8) ²⁵	119134.2 (2666.4) ²¹	– ⁰	122259.2 (1232.0)	123307.1 (814.5)
10	140457.4	135160.5 (1708.4)	135396.0 (1790.4)	134321.3 (1482.4) ²⁶	132616.3 (2328.0) ¹²	130967.4 (3191.0) ¹³	130994.8 (0.0) ¹	133567.5 (1820.9)	135642.4 (920.1)

Table S2: Data Set I, 750-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.0 (1.6)
3	42137.2	42137.2 (0.0)	42137.0 (1.0)	42137.0 (1.0)	42134.4 (4.1)	42134.2 (4.8)	42136.8 (1.5)	42137.2 (0.0)	42133.3 (5.0)
4	56182.9	56170.9 (10.1)	56170.9 (11.3)	56162.3 (13.6)	56153.8 (17.9)	56149.9 (14.0)	56146.9 (17.5)	56167.0 (13.8)	56139.3 (11.7)
5	70228.7	70169.6 (23.7)	70164.8 (25.1)	70150.9 (25.4)	70122.2 (26.3)	70118.1 (28.6)	70083.9 (31.7)	70125.8 (47.0)	70091.5 (35.7)
6	84274.4	84139.4 (24.3)	84130.1 (30.7)	84095.6 (36.9)	84035.2 (58.4)	84033.5 (58.3)	83948.2 (74.1)	84029.2 (141.3)	83965.7 (117.2)
7	98320.2	98064.7 (54.7)	98059.9 (46.4)	97910.8 (164.8)	97875.2 (169.3)	97840.0 (167.7)	97479.1 (285.1)	97774.4 (232.6)	97663.0 (196.3)
8	112365.9	111921.0 (119.4)	111924.1 (87.8)	111683.9 (282.2)	111506.5 (334.5)	111374.0 (460.8)	110595.8 (445.6)	111411.4 (493.5)	111287.8 (290.8)
9	126411.6	125552.7 (330.8)	125639.1 (260.7)	125207.2 (539.4)	124756.2 (610.5)	124698.0 (960.4)	121674.9 (2276.5) ²³	124780.0 (716.8)	124787.6 (424.2)
10	140457.4	139122.6 (345.7)	139212.9 (276.9)	138304.3 (685.9)	137817.6 (1160.1)	137748.6 (1116.3)	134937.7 (998.5) ⁴	138170.4 (764.7)	138182.1 (569.0)
11	154503.1	152343.5 (565.1)	152626.6 (443.6)	151286.3 (1117.4)	150404.1 (1042.2) ²⁹	149863.8 (1886.6) ²⁹	148555.7 (1134.8) ³	151004.2 (1339.0)	151173.5 (598.0)
12	168548.8	165526.4 (774.4)	165826.3 (766.5)	164279.2 (1284.1)	162903.5 (1685.8)	161630.5 (1988.2) ²⁸	– ⁰	163712.4 (1575.6)	164478.5 (634.3)

Table S3: Data Set I, 1000-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.3 (0.7)
3	42137.2	42137.2 (0.0)	42137.2 (0.0)	42137.2 (0.0)	42136.7 (1.6)	42136.7 (1.4)	42137.2 (0.0)	42137.2 (0.0)	42136.6 (1.8)
4	56182.9	56181.4 (3.1)	56181.8 (3.1)	56180.0 (3.8)	56169.4 (9.2)	56172.9 (7.8)	56166.9 (8.4)	56177.3 (8.6)	56165.3 (10.2)
5	70228.7	70209.8 (10.8)	70208.0 (13.1)	70202.1 (15.9)	70174.0 (19.7)	70183.3 (21.0)	70148.7 (22.4)	70193.3 (24.5)	70159.6 (39.9)
6	84274.4	84214.5 (19.7)	84213.1 (24.2)	84175.9 (36.0)	84149.9 (29.7)	84142.7 (23.5)	84096.1 (24.0)	84170.0 (49.5)	84064.6 (113.6)
7	98320.2	98192.1 (32.2)	98178.1 (38.0)	98128.4 (59.9)	98084.7 (47.0)	98078.4 (82.1)	97942.3 (67.9)	98064.6 (118.7)	97930.4 (157.1)
8	112365.9	112142.9 (48.9)	112125.1 (48.4)	112008.3 (131.2)	111947.1 (125.7)	111924.6 (164.7)	111608.3 (168.9)	111862.5 (246.1)	111736.9 (163.4)
9	126411.6	126003.5 (103.1)	125993.3 (89.8)	125842.0 (277.4)	125761.1 (168.4)	125701.5 (265.4)	124996.3 (282.9)	125589.7 (312.6)	125450.0 (305.1)
10	140457.4	139874.2 (102.1)	139834.3 (108.2)	139561.7 (359.4)	139348.0 (433.0)	139354.2 (388.5)	137969.0 (623.0)	139290.3 (447.7)	138991.7 (381.3)
11	154503.1	153621.0 (192.7)	153621.2 (168.9)	153184.6 (292.3)	152825.1 (385.8)	152736.8 (558.8)	150189.7 (1068.9) ²⁹	153010.5 (393.8)	152538.4 (405.6)
12	168548.8	167262.1 (383.9)	167223.9 (330.8)	166530.1 (760.0)	166091.9 (816.6)	165814.4 (827.9)	162390.4 (1765.9) ¹⁷	166510.6 (669.8)	166171.0 (462.2)
13	182594.6	180623.7 (496.4)	180638.4 (495.0)	179973.5 (642.1)	179101.5 (1028.9)	178913.6 (967.4)	174763.4 (1389.6) ⁴	179762.6 (1091.2)	179275.1 (644.9)
14	196640.3	194092.5 (813.0)	194082.6 (688.3)	193260.8 (939.8)	192235.7 (1504.8)	191046.6 (1754.6)	– ⁰	192874.6 (1745.6)	192657.7 (605.3)
15	210686.1	207308.1 (894.1)	207263.9 (720.0)	207119.3 (639.3)	204777.6 (1420.5)	203681.8 (1899.5) ²⁹	– ⁰	206486.7 (1093.6)	205876.8 (798.9)

S-III Per-Case Results and Additional Tables

Tables S1–S6 give the per-case results of the main comparison; the remaining tables give the minimum-spacing re-optimization, the full case-level statistics, the coordinates of the best layouts, the computational cost and the per-case values of the budget-split, initialization, budget, Horns Rev 1, IEA37 and robustness studies of the main text. All tables in this section are generated by `analysis/mpce_results.py`.

Table S4: Data Set II, 500-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14626.0 (5.6)
3	21946.1	21909.3 (15.9)	21904.5 (13.8)	21907.1 (18.9)	21900.9 (11.1)	21899.7 (9.5)	21901.2 (5.5)	21932.3 (17.2)	21892.0 (45.9)
4	29261.5	29124.8 (63.8)	29142.9 (54.2)	29067.5 (60.7)	29028.7 (93.8)	29004.2 (87.4)	29044.2 (61.5)	29070.8 (94.1)	28993.5 (96.1)
5	36576.9	36100.0 (100.0)	36118.0 (117.9)	35958.0 (130.6)	35918.9 (74.0)	35880.1 (168.6)	35789.6 (193.8)	35878.2 (160.9)	35914.9 (144.5)
6	43892.3	42851.5 (166.9)	42857.9 (163.5)	42638.7 (198.9)	42444.3 (224.5)	42512.1 (227.9)	42257.5 (374.9)	42560.3 (202.5)	42601.8 (178.6)
7	51207.6	49332.5 (195.5)	49337.3 (189.8)	49008.6 (239.7)	48887.9 (304.1)	48626.5 (340.5)	48203.9 (767.8) ¹⁵	49032.1 (314.4)	48959.7 (202.6)
8	58523.0	55727.9 (300.0)	55668.1 (312.6)	55274.9 (503.7)	54900.5 (471.3)	54699.5 (620.7) ²⁸	54451.3 (618.7) ²	55182.8 (738.6)	55427.1 (303.9)
9	65838.4	61754.3 (344.3)	61643.9 (481.7)	61380.0 (541.4)	60581.0 (844.8) ²⁵	59963.3 (652.2) ²¹	– ⁰	61166.8 (574.8)	61516.6 (326.2)
10	73153.8	67773.5 (477.6)	67472.2 (637.3)	67106.6 (779.5) ²⁵	65880.7 (761.6) ¹²	65868.7 (586.0) ¹³	64240.2 (0.0) ¹	67056.0 (749.1)	67297.9 (402.5) ²⁹

Table S5: Data Set II, 750-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.2 (1.8)
3	21946.1	21945.5 (2.5)	21945.0 (3.9)	21946.1 (0.0)	21940.0 (8.2)	21937.3 (8.1)	21942.5 (5.8)	21946.1 (0.0)	21938.3 (11.3)
4	29261.5	29225.9 (22.5)	29221.2 (22.7)	29210.5 (34.5)	29182.5 (36.2)	29201.4 (25.7)	29195.3 (26.2)	29227.3 (29.9)	29175.2 (48.2)
5	36576.9	36473.4 (36.9)	36480.6 (37.9)	36417.1 (67.3)	36332.0 (91.4)	36366.8 (74.3)	36277.7 (105.9)	36389.7 (97.1)	36267.4 (108.2)
6	43892.3	43596.3 (87.7)	43611.5 (96.3)	43500.1 (119.6)	43419.5 (153.1)	43377.4 (175.7)	43142.2 (144.8)	43367.4 (201.2)	43264.0 (140.5)
7	51207.6	50584.2 (207.2)	50592.0 (149.5)	50370.1 (169.6)	50264.8 (202.5)	50095.4 (193.6)	49696.6 (156.3)	50180.2 (303.8)	50174.3 (220.0)
8	58523.0	57411.1 (173.3)	57404.8 (185.9)	57033.9 (298.6)	56806.2 (303.6)	56659.2 (332.0)	56033.1 (331.4)	56855.9 (391.5)	56837.8 (340.9)
9	65838.4	64016.2 (259.2)	63991.1 (213.9)	63632.1 (366.8)	63291.6 (402.5)	63252.7 (342.8)	61970.8 (763.0) ²³	63217.0 (515.4)	63472.4 (272.8)
10	73153.8	70558.3 (205.2)	70484.8 (208.8)	70000.5 (393.6)	69520.3 (512.6)	69480.7 (385.7)	68103.2 (423.7) ⁴	69930.2 (355.0)	70128.5 (305.6)
11	80469.2	76912.3 (421.8)	76804.4 (395.8)	76265.0 (512.1)	75742.5 (677.1) ²⁹	75181.6 (681.1) ²⁹	74596.6 (476.3) ³	76099.7 (566.3)	76420.9 (338.9)
12	87784.5	82968.9 (469.9)	82874.2 (531.0)	82500.8 (492.0)	81822.1 (625.7)	81341.8 (792.1) ²⁸	– ⁰	82368.6 (603.1)	82792.2 (462.3)

Table S6: Data Set II, 1000-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.4 (1.2)
3	21946.1	21946.1 (0.0)	21946.1 (0.0)	21946.1 (0.0)	21945.6 (1.8)	21945.3 (2.8)	21946.1 (0.0)	21946.1 (0.0)	21943.9 (3.7)
4	29261.5	29254.5 (7.9)	29251.3 (10.2)	29247.6 (12.2)	29232.5 (18.5)	29228.0 (14.9)	29225.9 (12.1)	29253.1 (11.7)	29220.8 (35.1)
5	36576.9	36514.3 (28.7)	36519.7 (32.4)	36510.6 (35.3)	36485.9 (37.5)	36468.2 (46.1)	36402.8 (39.4)	36498.4 (49.1)	36426.0 (58.3)
6	43892.3	43776.8 (48.3)	43789.7 (42.7)	43736.8 (52.7)	43626.4 (74.5)	43656.6 (73.3)	43488.5 (69.7)	43686.1 (101.2)	43554.1 (85.6)
7	51207.6	50952.8 (83.2)	50969.1 (79.2)	50800.0 (120.6)	50731.8 (118.8)	50727.3 (119.3)	50442.7 (136.0)	50696.0 (203.8)	50596.6 (137.4)
8	58523.0	58050.4 (138.4)	58051.3 (154.0)	57883.4 (180.0)	57739.3 (142.0)	57645.1 (186.6)	57210.0 (191.8)	57588.6 (311.3)	57528.0 (226.3)
9	65838.4	65052.9 (170.2)	65065.3 (177.0)	64730.4 (233.0)	64476.4 (203.0)	64428.7 (273.1)	63790.1 (227.2)	64430.8 (311.5)	64357.7 (268.5)
10	73153.8	71906.7 (236.9)	71902.4 (238.0)	71541.9 (328.8)	71139.4 (336.8)	71099.4 (266.9)	70190.1 (286.9)	71342.3 (339.4)	71142.5 (257.2)
11	80469.2	78633.2 (289.7)	78478.9 (290.3)	78133.1 (465.6)	77697.5 (425.8)	77561.6 (498.0)	75960.9 (577.6) ²⁹	77948.1 (483.3)	77762.2 (436.7)
12	87784.5	85357.5 (330.7)	85172.5 (311.4)	84647.1 (650.8)	84049.6 (660.3)	83981.2 (515.0)	82211.6 (905.2) ¹⁷	84485.7 (606.8)	84385.5 (437.1)
13	95099.9	91904.0 (428.6)	91685.0 (453.5)	91179.4 (556.4)	90267.0 (482.6)	90303.5 (674.2)	88665.1 (496.0) ⁴	90787.8 (806.9)	90948.3 (429.2)
14	102415.3	98223.0 (410.2)	97889.6 (465.1)	97590.4 (615.0)	96936.5 (669.3)	96229.6 (1019.3)	– ⁰	97079.1 (980.3)	97371.5 (411.6)
15	109730.7	104794.8 (531.9)	104375.6 (501.6)	104035.3 (475.4)	102746.3 (827.8)	102327.4 (838.7) ²⁹	– ⁰	103777.6 (794.4)	103901.2 (406.2)

Table S7: Minimum-spacing re-optimization with the original LX-SSA and SSA code (6,030 calls, 30 seed-paired runs per cell): mean benchmark objective, Wilcoxon signed-rank p for LX-SSA vs. SSA, and the mean LX-SSA wake loss as a percentage of the ideal objective at 4D and 6D. A superscript gives the number of feasible runs when fewer than 30.

DS	r (m)	N	4D (308 m)			5D (385 m)			6D (462 m)			LX-SSA loss (%)
			LX-SSA	SSA	p	LX-SSA	SSA	p	LX-SSA	SSA	p	
1	500	4	56078.8	56086.8	0.237	56082.4	56085.2	0.700	56075.5	56083.9	0.839	0.19 $\rightarrow$ 0.19
1	750	8	111374.0	111506.5	0.245	111154.1	111163.4	0.871	110314.4 ²⁸	110847.5	0.021	0.88 $\rightarrow$ 1.85
1	1000	8	111924.6	111947.1	0.887	111953.5	111946.4	0.777	111781.5	111964.7	0.005	0.39 $\rightarrow$ 0.52
2	500	4	29004.2	29028.7	0.146	28999.5	29019.6	0.229	28981.8	28971.1	0.612	0.88 $\rightarrow$ 0.96
2	750	8	56659.2	56806.2	0.061	56621.0	56673.3	0.584	56533.9 ²⁸	56690.6	0.038	3.18 $\rightarrow$ 3.47
2	1000	8	57645.1	57739.3	0.084	57580.7	57655.7	0.140	57475.5	57587.2	0.114	1.50 $\rightarrow$ 1.79

Table S8: Largest turbine count for which a layout satisfying the circular boundary and the minimum spacing was constructed (multi-start packing search; constructive lower bound), compared with the largest tested N .

r (m)	4D (308 m)	5D (385 m)	6D (462 m)	Largest tested N
500	13	8	7	10
750	26	19	13	12
1000	42	29	21	15

Table S9: Case-level analysis over the 68 benchmark cases. Ranks of the mean feasible objective within each case (1 = best); a method with fewer than 15 feasible runs out of 30 in a case is ranked below all other methods, by its number of feasible runs (“<15”: number of such cases). Friedman $\chi^2_F = 319.3$ (7 d.f.), $p = 4.56 \times 10^{-65}$; Iman–Davenport $F_F = 136.5$, $p = 6.71 \times 10^{-109}$. p_z : Holm-adjusted p of the average-rank z test against PSO-VNS (best-ranked method: PSO-VNS). Because mean-rank post hoc tests depend on the pool of compared methods [8], pairwise two-sided Wilcoxon signed-rank tests on the per-case mean wake losses (%) are also given: p_W (Holm-adjusted over the 7 comparisons), matched-pairs rank-biserial correlation r_{rb} (positive = PSO-VNS better) and mean difference $\overline{\Delta L}$ of the wake loss (percentage points, PSO-VNS minus method, over the cases where both methods have at least 15 feasible runs). “Best”: cases in which the method alone ranks first; “Feas.”: percentage of feasible runs.

Method	Avg. rank	Best	Feas. (%)	<15	p_z	p_W	r_{rb}	$\overline{\Delta L}$ (pp)
PSO-VNS	1.82	29	100.0	0	–	–	–	–
PSO	1.97	25	100.0	0	0.713	0.296	+0.16	–0.018
SSA-VNS	3.45	0	99.6	0	2.04×10^{-4}	5.69×10^{-11}	+1.00	–0.330
VNS	4.40	3	100.0	0	2.42×10^{-9}	1.26×10^{-10}	+0.98	–0.465
SSA	5.38	0	97.6	2	8.32×10^{-17}	4.55×10^{-11}	+1.00	–0.611
MS-SLSQP	5.88	1	100.0	0	1.84×10^{-21}	4.18×10^{-11}	+0.96	–0.435
LX-SSA	6.06	0	96.9	2	3.33×10^{-23}	4.55×10^{-11}	+1.00	–0.765
DE	7.04	0	71.4	18	1.04×10^{-34}	5.59×10^{-11}	+1.00	–0.887

Table S10: Coordinates (m, farm center at the origin) of the best layout of all methods and of the best PSO-VNS layout for the largest N of each farm (6,030 calls); objective in benchmark units.

DS	r (m)	N	Method	Objective	Coordinates (x_i, y_i)
I	500	10	MS-SLSQP	138291.2	(469.2, 168.9); (127.1, -90.8); (-183.2, 21.4); (-365.2, -324.6); (-493.9, 32.0); (-320.1, 380.7); (273.4, 411.2); (-38.6, -356.1); (270.8, -412.5); (-6.8, 274.6)
I	500	10	PSO-VNS	137994.8	(485.3, -32.1); (-346.0, 302.7); (322.3, -382.2); (-376.7, -328.3); (317.9, 385.7); (142.8, 131.7); (-59.3, -101.1); (-46.5, 497.1); (-497.9, -0.5); (-88.7, -476.6)
I	750	12	PSO-VNS	167040.0	(-71.5, 110.7); (192.8, -653.4); (741.7, 104.8); (395.5, 126.3); (65.5, -302.0); (-728.9, 172.1); (185.8, 655.0); (-226.9, 453.5); (-531.3, -385.6); (534.1, 524.5); (542.8, -343.4); (-223.2, -715.4)
I	1000	15	PSO	208908.8	(-466.8, -48.4); (923.5, 23.2); (684.6, 693.4); (-230.6, 303.0); (524.3, -694.5); (137.7, 67.9); (517.2, 119.0); (137.7, 964.0); (153.5, -960.7); (-232.5, -607.3); (675.4, -366.5); (-684.6, -361.8); (-980.6, 75.4); (-48.5, -210.1); (-687.0, 528.7)
I	1000	15	PSO-VNS	208720.0	(594.1, 135.8); (-268.3, -387.2); (786.8, -517.5); (-79.4, 263.7); (960.7, 115.7); (-852.8, -274.3); (-998.2, 26.9); (104.5, -420.5); (-552.9, -78.4); (342.1, -881.7); (97.7, 888.8); (-84.5, -912.6); (-515.9, 610.4); (-707.3, 234.6); (355.1, 465.8)
II	500	10	PSO-VNS	68621.3	(499.3, -17.9); (-357.0, 305.5); (302.3, -322.3); (-362.6, -295.2); (336.9, 352.5); (46.0, 150.6); (-45.5, -150.2); (-65.8, 495.6); (-499.5, -2.6); (-102.9, -475.6)
II	750	12	PSO-VNS	83996.8	(-159.3, -214.1); (492.9, 564.5); (-391.5, 639.1); (121.1, -739.2); (-44.2, 619.1); (-662.7, 13.0); (-641.7, -309.1); (-465.3, 325.3); (-348.1, -575.4); (665.0, -320.3); (98.2, 250.2); (724.3, 15.3)
II	1000	15	PSO-VNS	105675.2	(-19.0, 118.7); (-84.9, -328.5); (-602.6, 187.3); (639.7, -523.0); (-942.8, -326.2); (216.5, -974.4); (928.9, -325.3); (-386.7, 906.8); (349.5, 906.0); (-310.8, 528.7); (993.7, 2.3); (322.4, -676.0); (469.3, 84.1); (701.3, 536.4); (-435.2, -798.8)

Table S11: Computational cost at 6,030 objective calls: mean wall-clock time per objective call (including the optimizer overhead) and range over N of the mean time per run (single core). Times come from the machine that produced each result file.

Method	ms per call	s per run
PSO-VNS	0.349	1.5–3.2
PSO	0.244	1.0–2.2
SSA-VNS	0.388	1.7–3.4
SSA	0.399	1.7–3.6
LX-SSA	0.396	1.7–3.6
DE	0.467	2.0–4.3
VNS	0.371	1.6–3.4
MS-SLSQP	1.820	2.6–21.7

Table S12: Sensitivity of PSO-VNS to the budget split between the PSO and VNS phases (25%, 50% and 75% of the 6,030 calls for PSO): mean benchmark objective of the feasible runs (superscript: feasible runs when fewer than 30) and Holm-adjusted Wilcoxon signed-rank p of the 50% split against the 25% and 75% splits (30 seed-paired runs).

DS	r	N	25%	50%	75%	p (25)	p (75)
I	500	6	83851.3	83918.5	83929.5	0.023	0.465
I	500	10	135131.6	135160.5	135421.8	0.556	0.069
I	750	8	111835.7	111921.0	111944.0	0.014	0.045
I	750	12	164894.0	165526.4	165991.2	0.004	0.004
I	1000	10	139700.6	139874.2	139883.7	0.002	0.529
I	1000	15	207367.9	207308.1	207724.0	0.570	3.20×10^{-5}
II	500	6	42809.4	42851.5	42883.5	0.110	0.110
II	500	10	67624.1	67773.5	67799.8	0.115	0.903
II	750	8	57358.0	57411.1	57448.4	0.121	0.121
II	750	12	82784.6	82968.9	83182.0	0.070	0.006
II	1000	10	71769.3	71906.7	71976.0	0.015	0.013
II	1000	15	104349.6	104794.8	104784.9	4.63×10^{-4}	0.968

Table S13: Horns Rev 1 site case, 16-turbine block (Vestas V80 power and thrust curves, measured 12-sector wind climate, installed outline; Jensen wake $k = 0.04$; minimum spacing $4D = 320$ m; wake-free AEP 149.26 GWh/yr): AEP in GWh/yr, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls, mean wake loss (%), feasible runs, Holm-adjusted Wilcoxon signed-rank p of PSO-VNS vs. each method and rank-biserial r_{rb} (positive = PSO-VNS better); run-level Friedman $p = 6.64 \times 10^{-28}$ (over the seeds common to all methods).

Layout / method	AEP	Loss (%)	Feas.	p_{Holm}	r_{rb}
Installed layout	139.51	6.53	–	–	–
PSO-VNS	138.73 (0.83)	7.05	30/30	–	–
SSA-VNS	137.98 (0.71)	7.56	30/30	6.67×10^{-4}	+0.68
LX-SSA-VNS	137.77 (0.91)	7.70	30/30	6.26×10^{-4}	+0.72
RS-VNS	137.96 (0.70)	7.57	28/30	2.24×10^{-4}	+0.78
VNS	137.67 (1.02)	7.77	30/30	2.24×10^{-4}	+0.78
PSO	138.06 (1.15)	7.51	19/30	1.38×10^{-5}	+0.88
SSA	136.66 (1.00)	8.44	30/30	2.79×10^{-7}	+0.97
LX-SSA	136.46 (0.90)	8.58	28/30	1.49×10^{-7}	+0.98
DE	–	–	0/30	1.68×10^{-8}	+1.00
MS-SLSQP	136.80 (0.77)	8.35	30/30	1.49×10^{-7}	+0.98

Table S14: Feasibility-preserving versus uniform random initialization at 6,030 calls on the six largest benchmark cases and the Horns Rev 16-turbine block (7 cases, 30 seed-paired runs each): mean wake loss (%) of the feasible runs averaged over the cases, percentage of feasible runs, and number of cases in which feasible initialization is significantly better / not different / worse (Wilcoxon signed-rank, Holm-adjusted over the methods of each case). Per-case values: Supplementary Table S15.

Method	Loss (%)		Feas. (%)		W/T/L
	random	feasible	random	feasible	
PSO-VNS	4.509	4.370	100.0	100.0	1/6/0

Table S15: Feasible versus random initialization per case (6,030 calls): mean wake loss (%; Horns Rev: mean AEP in GWh/yr) of the feasible runs, feasible runs, Holm-adjusted Wilcoxon signed-rank p (over the methods of the case) and rank-biserial r_{rb} (positive = feasible initialization better).

Method	Random	Feasible	Feas. (R)	Feas. (F)	p_{Holm}	r_{rb}
<i>Data Set I, $r = 500$ m, $N = 10$</i>						
PSO-VNS	3.771	2.871	30/30	30/30	0.004	+0.59
<i>Data Set I, $r = 750$ m, $N = 12$</i>						
PSO-VNS	1.793	1.909	30/30	30/30	0.339	-0.20
<i>Data Set I, $r = 1000$ m, $N = 15$</i>						
PSO-VNS	1.603	1.586	30/30	30/30	0.598	-0.11
<i>Data Set II, $r = 500$ m, $N = 10$</i>						
PSO-VNS	7.355	7.411	30/30	30/30	0.761	-0.07
<i>Data Set II, $r = 750$ m, $N = 12$</i>						
PSO-VNS	5.486	5.304	30/30	30/30	0.096	+0.35
<i>Data Set II, $r = 1000$ m, $N = 15$</i>						
PSO-VNS	4.498	4.549	30/30	30/30	0.556	-0.13
<i>Horns Rev 1, 16 turbines</i>						
PSO-VNS	138.73	138.88	30/30	30/30	0.318	+0.21

Table S16: Budget scaling (random initialization) on the six largest benchmark cases and the Horns Rev 16-turbine block: average rank over the 7 cases (ranking rule of Table I), mean wake loss (%) of the feasible runs averaged over the six benchmark cases in which the method has at least half of its runs feasible, and percentage of feasible runs. Per-case values: Table S17.

Method	6,030 calls			30,030 calls			120,030 calls		
	Rank	Loss	Feas.	Rank	Loss	Feas.	Rank	Loss	Feas.
PSO-VNS	1.43	4.084	100	1.14	3.336	100	2.71	3.126	100
PSO	1.86	4.181	95	3.00	3.512	99	4.43	3.268	100
SSA-VNS	4.14	4.678	96	4.71	3.725	100	4.71	3.209	100
SSA	8.29	4.828	83	8.57	4.767	97	8.86	4.202	100
LX-SSA	8.71	5.379	80	9.86	5.435	97	10.00	4.900	100
DE	10.00	–	1	5.71	3.930	82	3.43	3.286	100
VNS	6.43	4.950	100	3.86	3.724	100	5.00	3.284	100
MS-SLSQP	4.43	4.522	100	6.57	3.948	100	7.14	3.557	100
LX-SSA-VNS	5.14	4.802	99	5.57	3.896	100	4.71	3.178	100
RS-VNS	4.57	4.830	93	6.00	3.830	100	4.00	3.196	100

Table S17: Budget scaling per case: mean wake loss (%; Horns Rev: mean AEP in GWh/yr) of the feasible runs and feasible runs at each budget (random initialization).

Method	6,030	Feas.	30,030	Feas.	120,030	Feas.
<i>Data Set I, $r = 500$ m, $N = 10$ (wake loss, %)</i>						
PSO-VNS	3.771	30/30	2.506	30/30	2.261	22/22
PSO	3.604	30/30	2.994	20/20	2.705	7/7
SSA-VNS	4.369	26/30	2.848	20/20	2.202	8/8
SSA	5.582	12/30	4.836	18/20	4.110	8/8
LX-SSA	6.756	13/30	5.574	18/20	4.746	8/8
DE	6.737	1/30	3.139	20/20	3.014	7/7
VNS	4.905	30/30	2.961	20/20	2.420	7/7
MS-SLSQP	3.428	30/30	2.647	20/20	1.994	8/8
LX-SSA-VNS	4.888	29/30	3.556	20/20	2.080	8/8
RS-VNS	4.879	24/30	3.181	20/20	2.139	7/7
<i>Data Set I, $r = 750$ m, $N = 12$ (wake loss, %)</i>						
PSO-VNS	1.793	30/30	1.214	30/30	1.057	23/23
PSO	1.615	30/30	1.310	20/20	1.123	8/8
SSA-VNS	2.533	30/30	1.702	20/20	1.143	7/7
SSA	3.349	30/30	2.277	20/20	1.920	7/7
LX-SSA	4.105	28/30	2.451	20/20	2.206	8/8
DE	–	0/30	1.150	20/20	1.194	7/7
VNS	2.869	30/30	1.698	20/20	1.224	7/7
MS-SLSQP	2.415	30/30	1.895	20/20	1.549	7/7
LX-SSA-VNS	2.431	30/30	1.566	20/20	1.227	8/8
RS-VNS	2.404	30/30	1.718	20/20	1.172	8/8
<i>Data Set I, $r = 1000$ m, $N = 15$ (wake loss, %)</i>						
PSO-VNS	1.603	30/30	1.160	30/30	1.036	22/22
PSO	1.624	30/30	1.164	20/20	1.105	8/8
SSA-VNS	1.693	30/30	1.439	20/20	1.201	7/7
SSA	2.804	30/30	2.012	20/20	1.538	7/7
LX-SSA	3.325	29/30	2.622	20/20	2.074	7/7
DE	–	0/30	2.220	15/20	0.999	8/8
VNS	1.993	30/30	1.372	20/20	1.178	8/8
MS-SLSQP	2.283	30/30	1.902	20/20	1.646	7/7
LX-SSA-VNS	1.777	30/30	1.553	20/20	1.150	7/7
RS-VNS	1.946	30/30	1.529	20/20	1.108	8/8
<i>Data Set II, $r = 500$ m, $N = 10$ (wake loss, %)</i>						
PSO-VNS	7.355	30/30	6.750	30/30	6.391	23/23
PSO	7.767	30/30	7.006	20/20	6.311	7/7
SSA-VNS	8.266	25/30	7.112	20/20	6.418	8/8
SSA	9.942	12/30	8.325	18/20	7.708	8/8
LX-SSA	9.959	13/30	9.145	18/20	9.190	7/7
DE	12.185	1/30	6.920	20/20	6.821	8/8
VNS	8.336	30/30	6.981	20/20	6.681	8/8
MS-SLSQP	8.005	29/30	7.259	20/20	6.988	8/8
LX-SSA-VNS	8.548	29/30	7.185	20/20	6.556	7/7
RS-VNS	8.652	24/30	7.262	20/20	6.675	7/7
<i>Data Set II, $r = 750$ m, $N = 12$ (wake loss, %)</i>						
PSO-VNS	5.486	30/30	4.525	30/30	4.320	22/22
PSO	5.594	30/30	4.566	20/20	4.499	7/7
SSA-VNS	6.019	30/30	4.881	20/20	4.428	8/8
SSA	6.792	30/30	5.918	20/20	5.163	8/8
LX-SSA	7.339	28/30	6.642	20/20	5.700	8/8
DE	–	0/30	4.739	20/20	4.132	7/7
VNS	6.170	30/30	5.050	20/20	4.145	7/7
MS-SLSQP	5.687	30/30	5.129	20/20	4.789	8/8
LX-SSA-VNS	5.946	30/30	4.891	20/20	4.363	8/8
RS-VNS	5.888	30/30	4.934	20/20	4.399	7/7
<i>Data Set II, $r = 1000$ m, $N = 15$ (wake loss, %)</i>						
PSO-VNS	4.498	30/30	3.862	30/30	3.691	23/23
PSO	4.880	30/30	4.031	20/20	3.864	8/8
SSA-VNS	5.190	30/30	4.365	20/20	3.864	7/7
SSA	6.365	30/30	5.234	20/20	4.772	7/7
LX-SSA	6.747	29/30	6.178	20/20	5.483	8/8
DE	–	0/30	5.414	15/20	3.553	7/7
VNS	5.425	30/30	4.283	20/20	4.056	7/7
MS-SLSQP	5.313	30/30	4.854	20/20	4.374	7/7
LX-SSA-VNS	5.220	30/30	4.624	20/20	3.695	8/8
RS-VNS	5.210	30/30	4.354	20/20	3.684	8/8
<i>Horns Rev 1, 16 turbines (AEP, GWh/yr)</i>						
PSO-VNS	138.73	30/30	139.62	30/30	139.71	7/7
PSO	138.06	19/30	139.15	18/20	139.31	2/2
SSA-VNS	137.98	30/30	138.63	20/20	139.59	3/3
SSA	136.66	30/30	137.45	20/20	138.02	3/3
LX-SSA	136.46	28/30	136.74	20/20	136.92	3/3
DE	–	0/30	138.70	5/20	140.46	2/2
VNS	137.67	30/30	139.38	20/20	140.06	2/2
MS-SLSQP	136.80	30/30	137.77	20/20	138.35	3/3
LX-SSA-VNS	137.77	30/30	138.81	20/20	138.95	3/3
RS-VNS	137.96	28/30	138.76	20/20	139.55	2/2

Table S18: IEA Wind Task 37 Case Study 1 (16- and 36-turbine scenarios; official AEP model): AEP (MWh) mean, best and SD over the feasible runs, mean wake loss relative to the wake-free AEP (%), feasible runs, rank (ranking rule of Table I) and Holm-adjusted Wilcoxon signed-rank p of PSO-VNS vs. each method, per scenario and budget.

Method	Mean	Best	SD	Loss (%)	Feas.	Rank	p_{Holm}
<i>IEA37 CS1, 16 turbines, 6,030 calls</i>							
PSO-VNS	395,032	402,782	4,587	15.87	30/30	1	–
PSO	390,540	403,753	7,192	16.82	30/30	2	7.15×10^{-6}
SSA-VNS	388,563	403,251	8,564	17.25	30/30	3	3.80×10^{-4}
VNS	387,637	400,169	5,591	17.44	30/30	4	4.61×10^{-5}
LX-SSA-VNS	386,930	397,429	6,911	17.59	30/30	5	4.61×10^{-5}
RS-VNS	386,206	398,682	5,817	17.75	30/30	6	4.61×10^{-5}
MS-SLSQP	382,605	405,434	20,279	18.51	18/30	7	6.91×10^{-5}
SSA	378,073	395,150	9,076	19.48	30/30	8	6.52×10^{-8}
LX-SSA	373,977	393,285	7,867	20.35	30/30	9	1.68×10^{-8}
DE	338,611	357,391	8,837	27.88	27/30	10	1.68×10^{-8}
<i>IEA37 CS1, 16 turbines, 30,030 calls</i>							
PSO-VNS	402,583	413,332	4,678	14.26	30/30	1	–
PSO	401,347	412,702	4,764	14.52	30/30	2	0.006
LX-SSA-VNS	399,197	407,330	4,330	14.98	30/30	3	0.006
SSA-VNS	396,921	406,925	6,267	15.47	30/30	4	4.42×10^{-4}
VNS	396,192	404,076	4,928	15.62	30/30	5	7.99×10^{-5}
RS-VNS	396,163	408,205	4,977	15.63	30/30	6	8.52×10^{-6}
MS-SLSQP	394,567	410,350	15,316	15.97	29/30	7	0.006
SSA	392,876	401,459	4,394	16.33	30/30	8	4.30×10^{-7}
LX-SSA	390,267	398,770	5,052	16.88	30/30	9	1.68×10^{-8}
DE	359,952	397,403	12,352	23.34	30/30	10	2.98×10^{-8}
<i>IEA37 CS1, 36 turbines, 6,030 calls</i>							
VNS	741,566	785,791	24,100	29.81	30/30	1	0.404
PSO-VNS	736,832	787,709	19,268	30.25	30/30	2	–
PSO	729,189	787,989	21,150	30.98	27/30	3	8.49×10^{-4}
SSA-VNS	728,870	766,803	22,859	31.01	30/30	4	0.248
SSA	717,258	750,707	19,300	32.11	30/30	5	0.001
LX-SSA-VNS	715,952	757,838	30,626	32.23	29/30	6	0.002
LX-SSA	709,661	746,301	19,845	32.83	30/30	7	4.75×10^{-4}
RS-VNS	709,276	750,170	21,667	32.86	28/30	8	3.60×10^{-5}
MS-SLSQP	733,342	748,571	13,798	30.58	4/30	9	1.49×10^{-7}
DE	–	–	–	–	0/30	10	1.68×10^{-8}
<i>IEA37 CS1, 36 turbines, 30,030 calls</i>							
PSO-VNS	807,446	827,385	11,697	23.57	30/30	1	–
PSO	802,252	830,256	13,260	24.06	30/30	2	4.87×10^{-5}
VNS	801,797	824,230	11,876	24.10	30/30	3	0.119
SSA-VNS	791,404	819,931	14,356	25.09	30/30	4	4.18×10^{-4}
RS-VNS	787,892	807,208	9,532	25.42	30/30	5	4.16×10^{-6}
LX-SSA-VNS	785,677	810,989	13,216	25.63	30/30	6	1.12×10^{-7}
SSA	771,298	802,538	14,756	26.99	30/30	7	2.98×10^{-8}
LX-SSA	752,030	789,311	18,838	28.82	30/30	8	2.98×10^{-8}
MS-SLSQP	798,913	833,365	41,511	24.38	14/30	9	2.37×10^{-4}
DE	–	–	–	–	0/30	10	1.68×10^{-8}

Table S19: IEA Wind Task 37 Case Study 1: AEP (MWh, official calculator) of the baseline (example) layout, of the best feasible and the best overall participant layout, our best feasible layout over all methods and budgets, the best PSO-VNS layout, mean PSO-VNS AEP (budgets 6,030 / 30,030 calls), and difference (%) of the best PSO-VNS layout from the best feasible participant layout and from the baseline. “infeas.”: violates the boundary or spacing constraint by more than 1 mm.

Scenario	Baseline	Best feasible publ.	Best publ.	Our best	PSO-VNS best	PSO-VNS mean	Δ_{publ}	Δ_{base}
IEA37 CS1, 16 turbines	366,941.6	418,924.4 (par4)	421,561.9 (par12, infeas.)	413,331.9 (PSO-VNS)	413,331.9	395,031.8 / 402,583.1	-1.33	+12.64
IEA37 CS1, 36 turbines	737,883.1	863,676.3 (par4)	882,383.3 (par12, infeas.)	833,365.0 (MS-SLSQP)	827,385.0	736,832.4 / 807,446.0	-4.20	+12.13

Table S20: Robustness of the results to the benchmark model. All feasible final layouts of the 8 methods are re-evaluated (not re-optimized) with alternative power-curve and wake models. Δ : mean relative change of the objective (expected power) with respect to the benchmark model (both evaluated on the stored coordinates, rounded to 1 mm); average rank of each method over the 68 cases (ranking rule of Table I: methods with fewer than half of their runs feasible are ranked last, by their number of feasible runs); $\bar{\tau}$: mean Kendall rank correlation between the benchmark and the alternative ordering of the methods with at least 15 feasible runs within a case; “Same best”: percentage of cases in which the best method is unchanged.

Model	Δ (%)		Average rank										$\bar{\tau}$	Same best (%)
	DS I	DS II	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP				
Benchmark (linear curve, Jensen)	+0.0	+0.0	1.82	1.97	3.45	5.38	6.06	7.04	4.40	5.88	–	–		
Cubic power curve	-17.7	-36.9	1.82	1.90	3.53	5.68	5.97	6.90	4.42	5.79	0.95	94		
Cubic curve with 25 m/s cut-out	-21.5	-37.0	1.80	1.91	3.53	5.68	5.97	6.90	4.42	5.79	0.94	90		
Gaussian wake ($k^* = 0.04$)	-0.3	-0.4	1.97	1.91	3.93	5.43	6.04	6.94	5.12	4.66	0.66	54		

S-IV Wind Data

Table S21: Wind Data Sets I and II: probability p_l of direction bin l ($[15(l-1), 15l)^\circ$, direction the wind blows towards) and Weibull scale ψ ; $\zeta = 2$ in all bins.

Bin l	1	2	3	4	5	6	7	8	9	10	11	12
DS I: p_l ($\psi = 13$)	0.000	0.010	0.010	0.010	0.010	0.200	0.600	0.010	0.010	0.010	0.010	0.010
DS II: ψ	7.0	5.0	5.0	5.0	5.0	4.0	5.0	6.0	7.0	7.0	8.0	9.5
DS II: p_l	0.0002	0.0080	0.0227	0.0242	0.0225	0.0339	0.0423	0.0290	0.0617	0.0813	0.0994	0.1394
Bin l	13	14	15	16	17	18	19	20	21	22	23	24
DS I: p_l ($\psi = 13$)	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.000
DS II: ψ	10.0	8.5	8.5	6.5	4.6	2.6	8.0	5.0	6.4	5.2	4.5	3.9
DS II: p_l	0.1839	0.1115	0.0765	0.0080	0.0051	0.0019	0.0012	0.0010	0.0017	0.0031	0.0097	0.0317

S-V Comparison with Published Kusiak–Song Results

The Kusiak–Song benchmark has been solved with an evolution strategy [1], ant colony optimization [2], particle filtering [3] and biogeography-based optimization [4]. [TBD: author: literature table pending verification of the published values against the original papers (see analysis/literature_values.md)]

S-VI Additional Figures

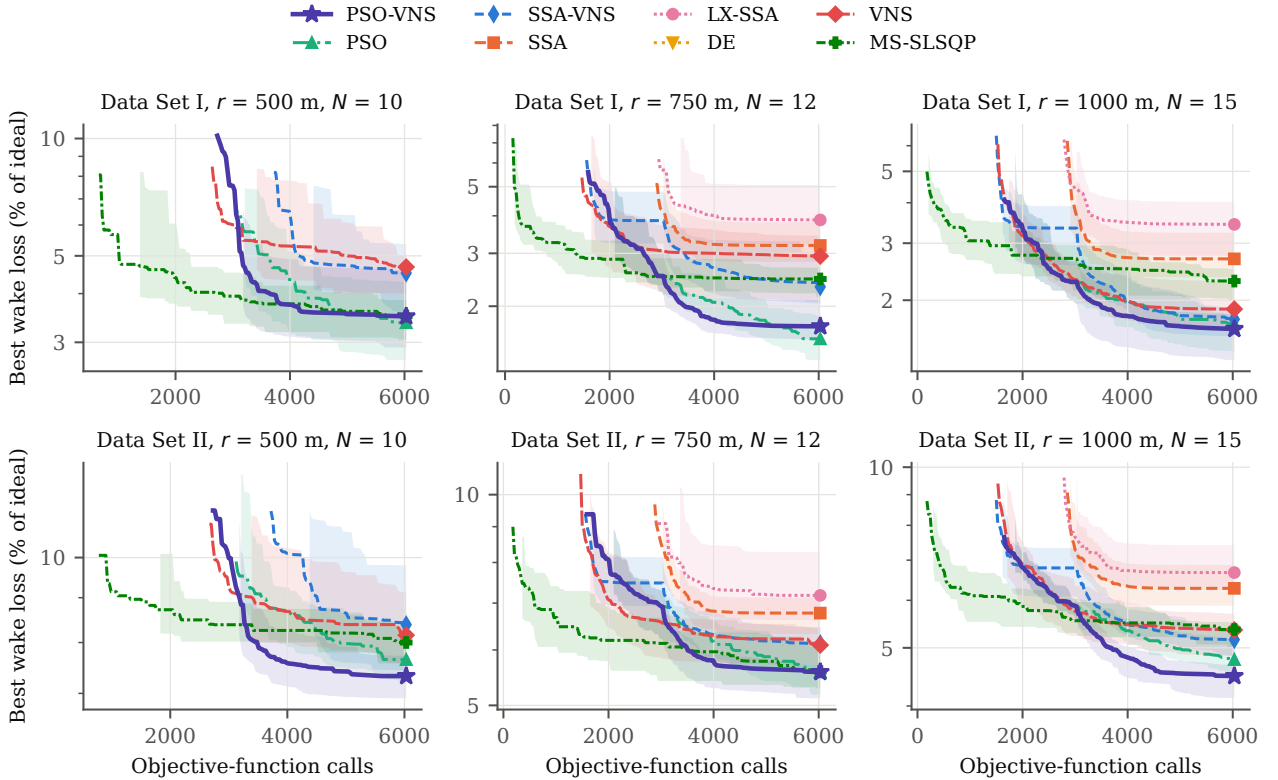


Figure S4: Convergence of the eight methods of the main comparison for the largest N of each farm: median (line) and interquartile range (band) of the best feasible wake loss over 30 runs versus objective calls.

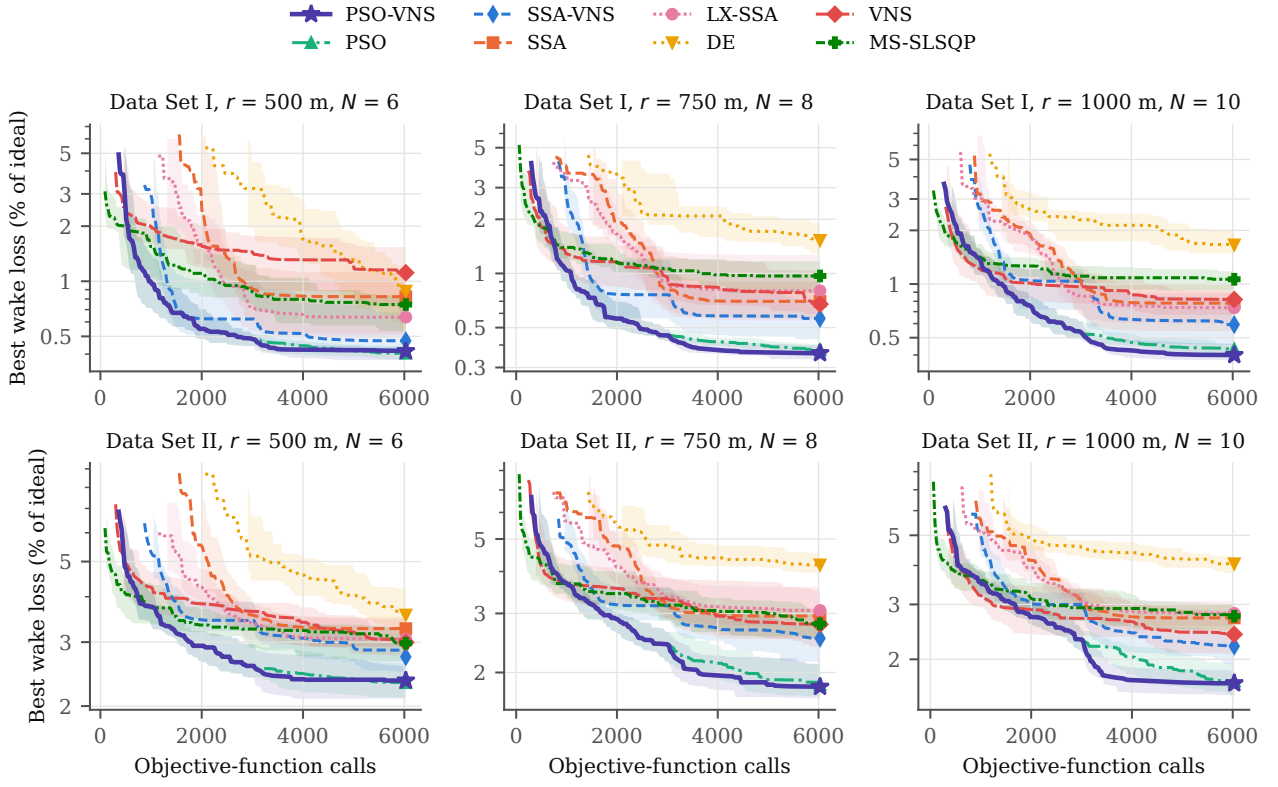


Figure S5: As Fig. S4, for the middle N of each farm ($N = 6, 8$ and 10 for $r = 500, 750$ and 1000 m).

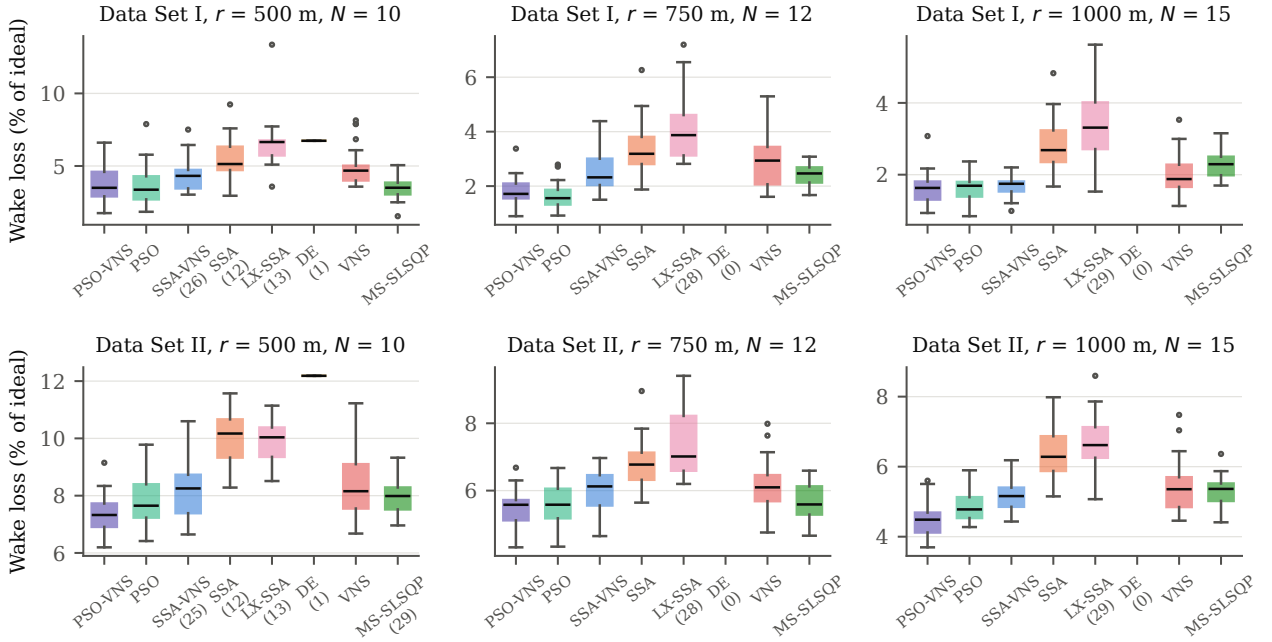


Figure S6: Distribution of the wake loss of the feasible runs for the largest N of each farm (number of feasible runs in parentheses when fewer than 30).

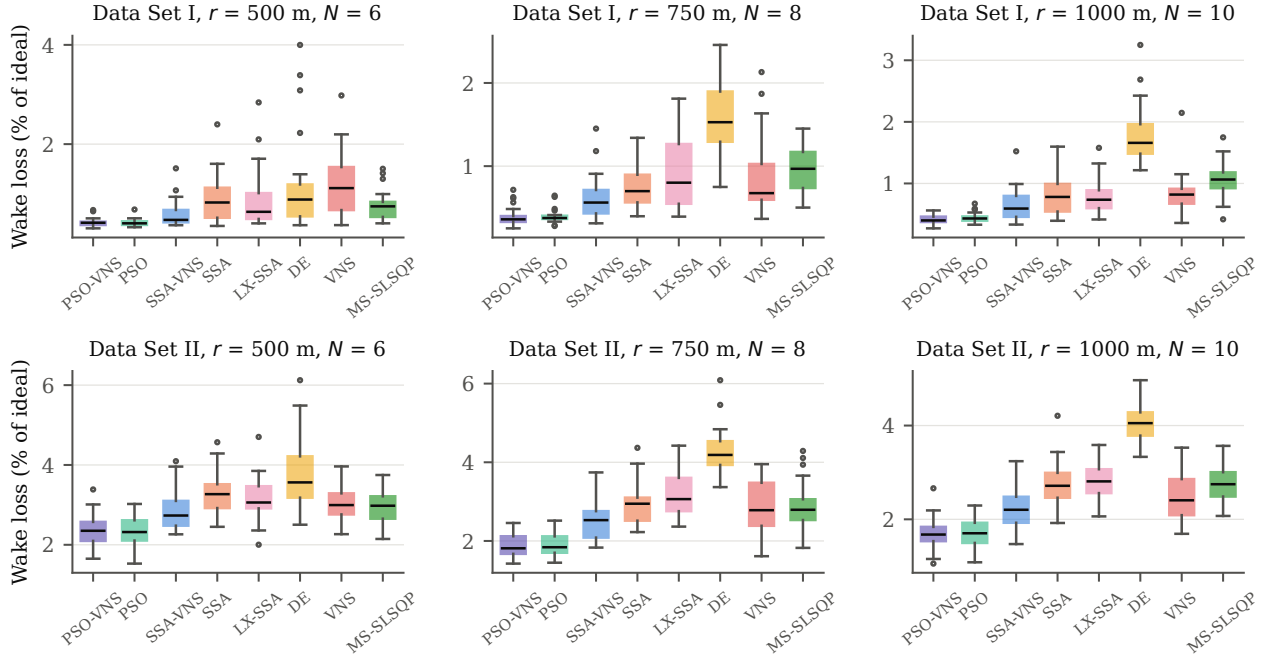


Figure S7: As Fig. S6, for the middle N of each farm.

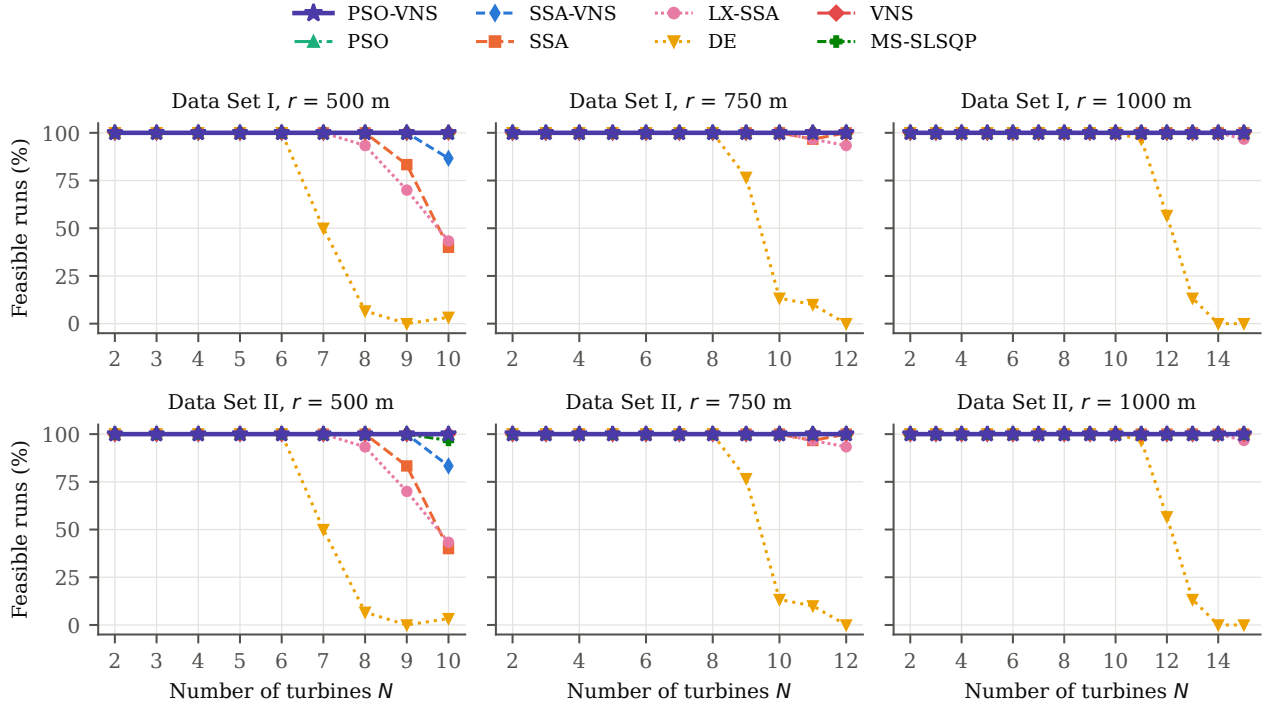


Figure S8: Percentage of feasible runs versus the number of turbines.

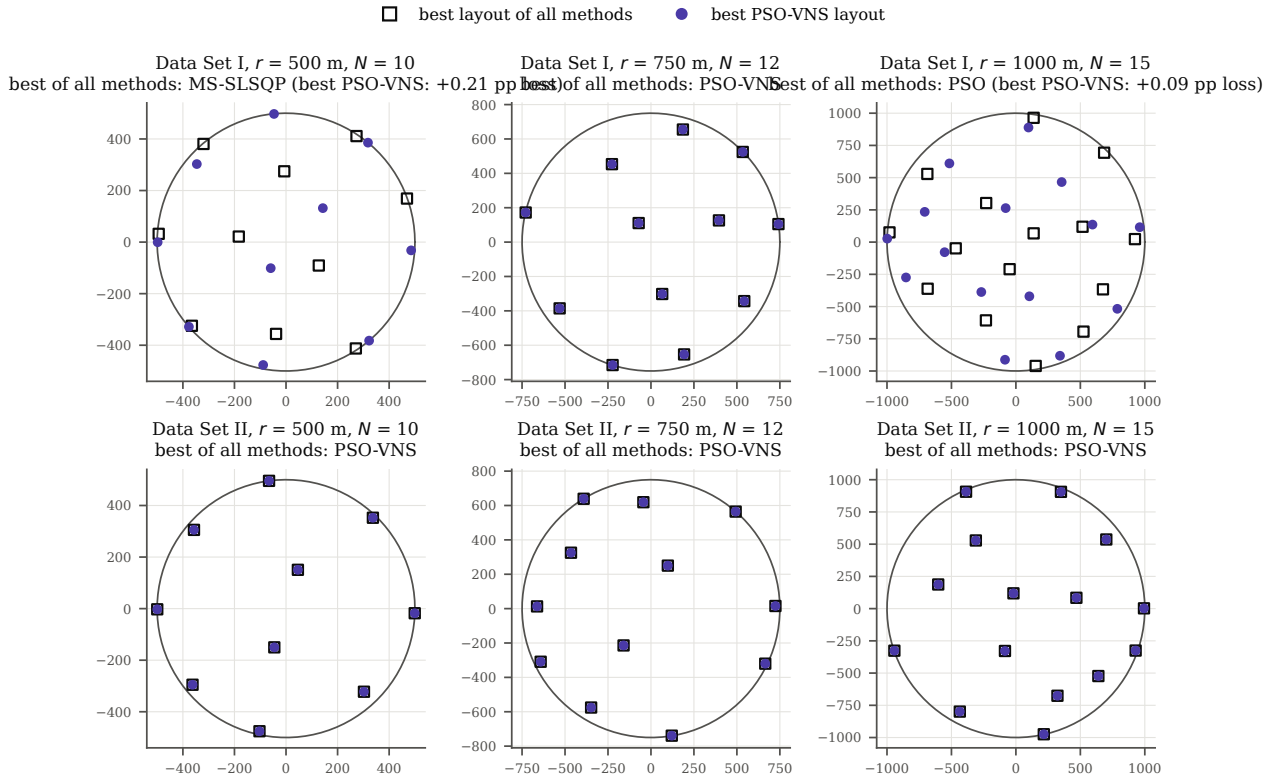


Figure S9: Best layout of all methods (squares) and best PSO-VNS layout (dots) for the largest N of each farm; the coordinates are listed in Table S10.

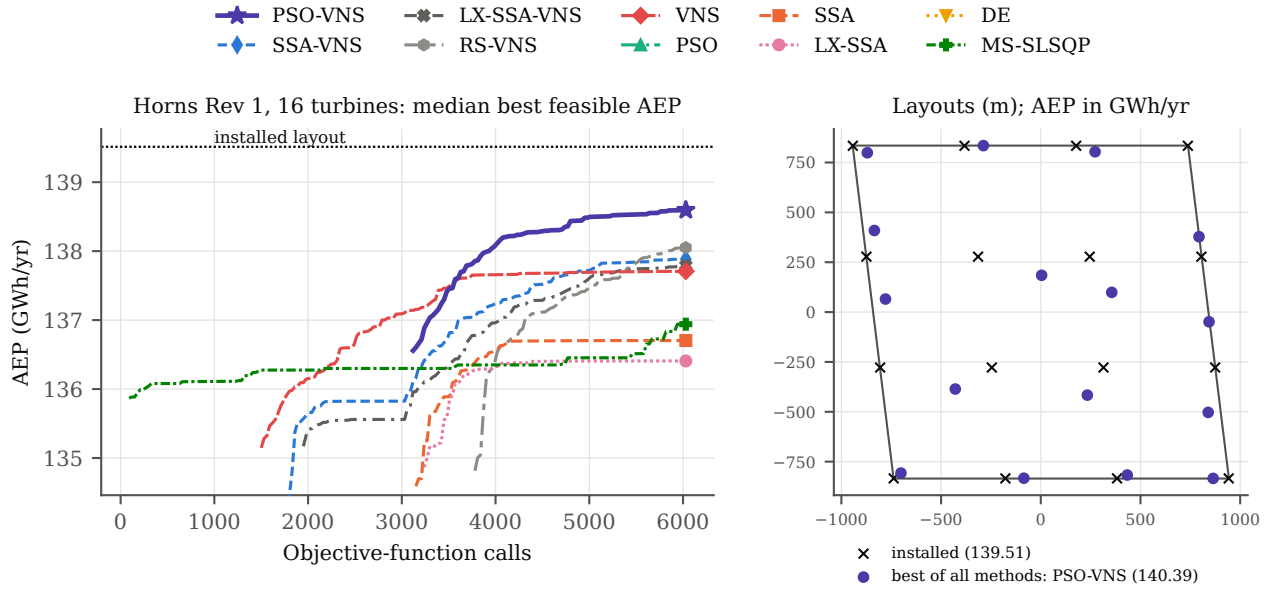


Figure S10: Horns Rev 1 16-turbine block: median best feasible AEP versus objective calls (left) and layouts (right).

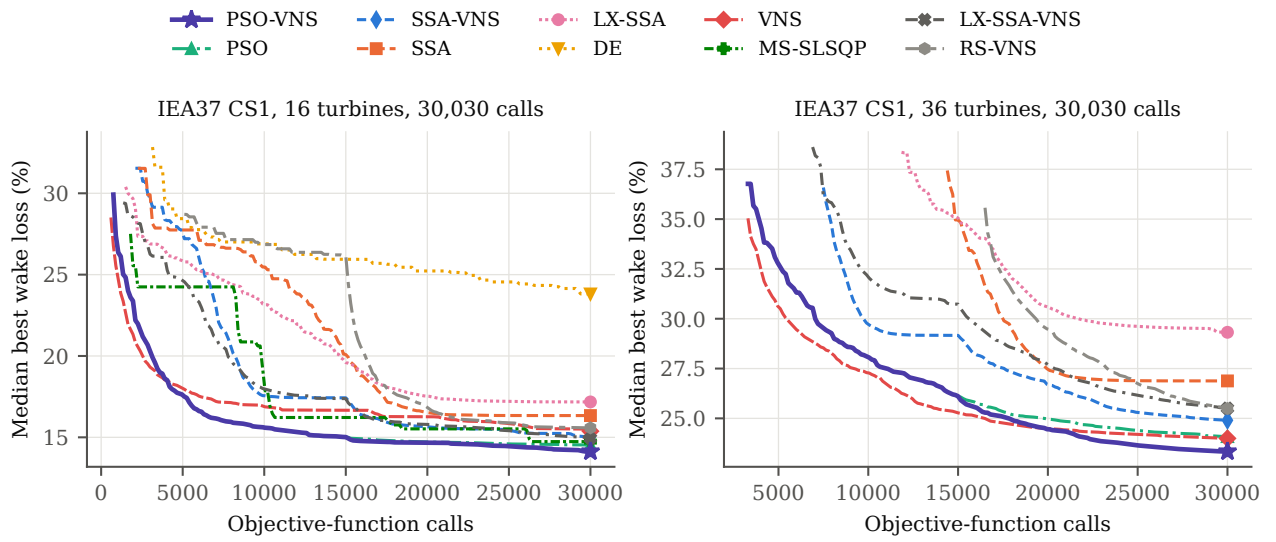


Figure S11: IEA37 Case Study 1: median best feasible wake loss versus objective calls at the largest budget.

References

- [1] A. Kusiak and Z. Song, “Design of wind farm layout for maximum wind energy capture,” *Renew. Energy*, vol. 35, no. 3, pp. 685–694, 2010.
- [2] Y. Eroğlu and S. U. Seçkiner, “Design of wind farm layout using ant colony algorithm,” *Renew. Energy*, vol. 44, pp. 53–62, 2012.
- [3] Y. Eroğlu and S. U. Seçkiner, “Wind farm layout optimization using particle filtering approach,” *Renew. Energy*, vol. 58, pp. 95–107, 2013.
- [4] J. C. Bansal and P. Farswan, “Wind farm layout using biogeography based optimization,” *Renew. Energy*, vol. 107, pp. 386–402, 2017.
- [5] S. Mirjalili *et al.*, “Salp swarm algorithm: A bio-inspired optimizer for engineering design problems,” *Adv. Eng. Softw.*, vol. 114, pp. 163–191, 2017.
- [6] P. Solanki and K. Deep, “Laplacian salp swarm algorithm for continuous optimization,” *Int. J. Syst. Assur. Eng. Manag.*, 2023.
- [7] M. Bastankhah and F. Porté-Agel, “A new analytical model for wind-turbine wakes,” *Renew. Energy*, vol. 70, pp. 116–123, 2014.
- [8] A. Benavoli, G. Corani, and F. Mangili, “Should we really use post-hoc tests based on mean-ranks?” *J. Mach. Learn. Res.*, vol. 17, no. 5, pp. 1–10, 2016.
- [9] N. F. Baker *et al.*, “Best practices for wake model and optimization algorithm selection in wind farm layout optimization,” in *Proc. AIAA SciTech Forum*, 2019, Paper AIAA 2019-0540.
- [10] N. F. Baker *et al.*, “IEA Wind Task 37 wind farm layout optimization case studies: Case files and submitted results.” [Online]. Available: <https://github.com/IEAWindTask37/iea37-wflo-casestudies>